

# VALIDATION PLAN

## OPERATIONAL QUALIFICATION

### SHELF LIFE & DEGRADATION ANALYSIS MODULE

**JR Validated Environment — Shelf Life & Degradation Analysis Module v2.0**

| Field            | Value                                                                                 |
|------------------|---------------------------------------------------------------------------------------|
| Document Number  | JR-VP-SHELF-001                                                                       |
| Title            | Validation Plan — Operational Qualification, Shelf Life & Degradation Analysis Module |
| System           | JR Validated Environment — Shelf Life Module                                          |
| Module Version   | 2.0                                                                                   |
| Document Version | 2.0                                                                                   |
| Status           | Approved                                                                              |
| Effective Date   | 2026-05-18                                                                            |
| Author           | Joep Rous                                                                             |
| Reviewer         | [Name]                                                                                |
| Approver         | [Name]                                                                                |

Version History

| Version | Date       | Author    | Description                                                                                                                                               |
|---------|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 2026-04-19 | Joep Rous | Initial release. Covers all 5 Shelf Life scripts in module v1.0. 57 test cases including --transform log (TC-SHELF-LIN-013..016).                         |
| 2.0     | 2026-05-18 | Joep Rous | Added --report DV report TCs for all 5 scripts: Q10-011..013, ARR-011..013, LIN-017..019, POOL-011..013, EXT-012..014. Added UR-SHELF-028. Total TCs: 72. |

Approval Signatures

| Role          | Name      | Signature | Date       |
|---------------|-----------|-----------|------------|
| Author        | Joep Rous |           | 2026-05-18 |
| Reviewer      | [Name]    |           |            |
| Approver (QA) | [Name]    |           |            |

## 1. Purpose

This document defines the Validation Plan for the Operational Qualification (OQ) of the Shelf Life & Degradation Analysis module included in version 1.0 of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability between requirements and tests needed to demonstrate that each Shelf Life script performs correctly for its intended use.

Execution of the tests defined in this plan, together with satisfactory results and documented evidence, constitutes the OQ for the Shelf Life module.

## 2. Scope

### 2.1 In Scope

- All 5 R scripts in repos/shelf\_life/R/ (jrc\_shelf\_life\_q10, jrc\_shelf\_life\_arrhenius, jrc\_shelf\_life\_linear, jrc\_shelf\_life\_poolability, jrc\_shelf\_life\_extrapolate)
- Correct computation of acceleration factor (AF) and real-time equivalent via Q10 method (ASTM F1980) and Arrhenius kinetics (ISO 11607 / ICH Q1E)
- Correct linear degradation modelling: Brown-Forsythe homogeneity test, OLS regression, shelf life estimation as intersection of CI of predicted mean with specification limit
- Correct --transform log behaviour: log-scale OLS with CI back-transformed via exp() to original measurement scale
- Correct ICH Q1E batch poolability analysis: two-step ANCOVA at  $\alpha=0.25$ , FULL POOL / PARTIAL POOL / DO NOT POOL decision
- Correct extrapolation of CI from a saved model CSV: ICH Q1E 50% warning and 100% hard stop; transform field read and applied correctly
- Correct rejection of invalid inputs with informative error messages
- Correct file I/O behaviour (CSV input, PNG and model CSV output to ~/Downloads/)
- Verification that all R scripts enforce execution within the jrrun validated environment by checking RENV\_PATHS\_ROOT at startup

### 2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by IQ; see JR-VP-001)
- Performance qualification (PQ) — real product stability studies are conducted separately under individual stability protocols
- User acceptance testing (UAT)
- Validation of R language interpreter and third-party packages (ggplot2, nlme) — package integrity is confirmed at installation via SHA256 (JR-IQ-001)
- Statistical validation of the underlying methods (Arrhenius kinetics, ICH Q1E poolability decision rules, OLS inference) — method references are cited in Section 3

## 3. Regulatory Basis and References

### 3.1 Regulations

| Reference | Title |
|-----------|-------|
|-----------|-------|

|                             |                                                              |
|-----------------------------|--------------------------------------------------------------|
| 21 CFR Part 820, §820.70(i) | Quality System Regulation — Automated Data Processing        |
| 21 CFR Part 820, §820.75    | Process Validation                                           |
| 21 CFR Part 11              | Electronic Records; Electronic Signatures                    |
| ISO 13485:2016, §7.5.6      | Validation of processes for production and service provision |

## 3.2 FDA Guidance

| Reference | Title                                                                                  | Date           |
|-----------|----------------------------------------------------------------------------------------|----------------|
| FDA GPSV  | General Principles of Software Validation; Final Guidance for Industry and FDA Staff   | January 2002   |
| FDA CSA   | Computer Software Assurance for Production and Quality System Software; Draft Guidance | September 2022 |

## 3.3 Standards and Methods

| Reference               | Title                                                                                                                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM F1980-16           | Standard Guide for Accelerated Aging of Sterile Barrier Systems for Medical Devices. Q10 method basis. $AF = Q10^{(\Delta T/10)}$ .                                                                                                                                                                          |
| ISO 11607-1:2019        | Packaging for terminally sterilized medical devices — Part 1: Requirements for materials, sterile barrier systems and packaging systems. References Arrhenius method for accelerated ageing.                                                                                                                 |
| ICH Q1E                 | Evaluation for Stability Data. International Council for Harmonisation. Poolability decision rules (two-step ANCOVA at $\alpha=0.25$ ); extrapolation limits (50%/100% beyond last observation).                                                                                                             |
| ICH Q1A(R2)             | Stability Testing of New Drug Substances and Products. Defines shelf life estimation from stability regression and CI of the mean.                                                                                                                                                                           |
| ISO 13485:2016          | Medical devices — Quality management systems — Requirements for regulatory purposes.                                                                                                                                                                                                                         |
| Arrhenius (1889)        | Arrhenius SA. Über die Reaktionsgeschwindigkeit bei der Inversion von Rohrzucker durch Säuren. Zeitschrift für physikalische Chemie. $AF = \exp(E_a/R \times (1/T_{real} - 1/T_{accel}))$ ; $R = 1.987 \times 10^{-3} \text{ kcal}/(\text{mol} \cdot \text{K})$ .                                            |
| Brown & Forsythe (1974) | Robust tests for equality of variances. Journal of the American Statistical Association, 69(346), 364–367. Implemented as one-way ANOVA on absolute deviations from group medians using base R <code>lm()</code> and <code>anova()</code> — equivalent to <code>car::leveneTest(..., center=median)</code> . |

## 3.4 Internal Documents

| Document Number | Title                                                                         |
|-----------------|-------------------------------------------------------------------------------|
| JR-VP-001       | Validation Plan — Installation Qualification, JR Validated Environment v1.0.0 |
| JR-IQ-001       | IQ Execution Evidence (docs/IQ_validation_*.txt)                              |
| JR-VP-002       | Validation Plan — OQ, Community Script Suite                                  |

## 4. Definitions and Abbreviations

| Term             | Definition                                                                                                                                                         |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AF               | Acceleration Factor — ratio of real-time to accelerated ageing duration that achieves equivalent degradation.                                                      |
| Q10              | Temperature coefficient — the factor by which reaction rate increases per 10°C rise. Typically 1.5–3.0 for medical device degradation.                             |
| E <sub>a</sub>   | Activation energy (kcal/mol) — Arrhenius model parameter. Typical range for medical device degradation: 10–25 kcal/mol.                                            |
| R (gas constant) | $1.987 \times 10^{-3} \text{ kcal}/(\text{mol} \cdot \text{K})$ — used in the Arrhenius equation.                                                                  |
| ICH Q1E          | International Council for Harmonisation guideline on evaluation for stability data. Defines poolability analysis, shelf life estimation, and extrapolation limits. |
| ANCOVA           | Analysis of Covariance. Used in ICH Q1E poolability: tests batch×time interaction (Step 1) and batch main effect (Step 2) at $\alpha=0.25$ .                       |

|                 |                                                                                                                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brown-Forsythe  | Homogeneity-of-variance test robust to non-normal data. Implemented as one-way ANOVA on absolute deviations from group medians (base R only — no car package).                                                                                            |
| OLS             | Ordinary Least Squares regression. Used in <code>jrc_shelf_life_linear</code> to fit <code>value ~ time</code> (or <code>log(value) ~ time</code> with <code>--transform log</code> ).                                                                    |
| CI of the mean  | Confidence interval for the predicted mean response. Shelf life is estimated as the time at which the CI bound of the predicted mean crosses the specification limit (ICH Q1E approach, not prediction interval).                                         |
| Transform (log) | <code>--transform log</code> flag in <code>jrc_shelf_life_linear</code> . Fits <code>lm(log(value) ~ time)</code> ; CI is back-transformed via <code>exp()</code> to the original measurement scale. Use for right-skewed data or log-normal degradation. |
| FULL POOL       | ICH Q1E poolability decision: slopes and intercepts of all batches are not statistically different (Step 1 $p > 0.25$ and Step 2 $p > 0.25$ ). A single regression line may be used for all batches.                                                      |
| PARTIAL POOL    | ICH Q1E poolability decision: slopes are not statistically different (Step 1 $p > 0.25$ ) but intercepts differ (Step 2 $p \leq 0.25$ ). Use the batch with the shortest shelf life.                                                                      |
| DO NOT POOL     | ICH Q1E poolability decision: slopes differ significantly (Step 1 $p \leq 0.25$ ). Separate regression lines must be used; the batch with the shortest shelf life applies.                                                                                |
| IQ              | Installation Qualification                                                                                                                                                                                                                                |
| OQ              | Operational Qualification                                                                                                                                                                                                                                 |
| PQ              | Performance Qualification                                                                                                                                                                                                                                 |
| UR              | User Requirement                                                                                                                                                                                                                                          |
| TC              | Test Case                                                                                                                                                                                                                                                 |
| jrrun           | The validated bash entry-point script that sets the controlled environment before executing any script                                                                                                                                                    |
| CSV             | Comma-separated values input file                                                                                                                                                                                                                         |
| PNG             | Portable Network Graphics output file; plots saved to <code>~/Downloads/</code>                                                                                                                                                                           |

## 5. User Requirements

The following user requirements define the expected behaviour of the Shelf Life & Degradation Analysis module. Each requirement is verifiable by the test cases in Section 7.

| UR ID        | Title                                            | Requirement                                                                                                                                                                                                                                               |
|--------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UR-SHELF-001 | Bypass protection — <code>RENV_PATHS_ROOT</code> | All Shelf Life scripts shall check the <code>RENV_PATHS_ROOT</code> environment variable at startup and call <code>stop()</code> with an informative message if it is not set, preventing execution outside the validated <code>jrrun</code> environment. |
| UR-SHELF-002 | Error handling — non-zero exit                   | All Shelf Life scripts shall exit with a non-zero status code whenever an error condition is detected, and shall print an informative message.                                                                                                            |
| UR-SHELF-003 | Q10 — acceleration factor computation            | <code>jrc_shelf_life_q10</code> shall compute the acceleration factor as $AF = Q10^{((accel\_temp - real\_temp) / 10)}$ and the real-time equivalent as $real\_time = accel\_time \times AF$ (ASTM F1980).                                                |
| UR-SHELF-004 | Q10 — sensitivity table                          | <code>jrc_shelf_life_q10</code> shall report a sensitivity table showing AF and real-time equivalent for $Q10 - 0.5$ , $Q10$ (stated), and $Q10 + 0.5$ .                                                                                                  |
| UR-SHELF-005 | Q10 — range warning                              | <code>jrc_shelf_life_q10</code> shall emit a warning when $Q10$ is outside the typical range $[1.5, 3.0]$ and continue with exit 0.                                                                                                                       |
| UR-SHELF-006 | Q10 — temperature direction check                | <code>jrc_shelf_life_q10</code> shall reject inputs where $accel\_temp \leq real\_temp$ with a non-zero exit, because no acceleration occurs.                                                                                                             |
| UR-SHELF-007 | Arrhenius — acceleration factor computation      | <code>jrc_shelf_life_arrhenius</code> shall compute $AF = \exp(Ea/R \times (1/T_{real} - 1/T_{accel}))$ where $R = 1.987 \times 10^{-3}$ kcal/(mol·K) and temperatures are in Kelvin (ISO 11607 / ICH Q1E).                                               |
| UR-SHELF-008 | Arrhenius — unit flag                            | <code>jrc_shelf_life_arrhenius</code> shall accept <code>--unit C</code> (Celsius, default) or <code>--unit K</code> (Kelvin) and produce identical AF values when equivalent temperatures are supplied in either unit.                                   |
| UR-SHELF-009 | Arrhenius — sensitivity table                    | <code>jrc_shelf_life_arrhenius</code> shall report a sensitivity table showing AF and real-time equivalent for $Ea - 2$ , $Ea$ (stated), and $Ea + 2$ kcal/mol.                                                                                           |
| UR-SHELF-010 | Arrhenius — $Ea$ range warning                   | <code>jrc_shelf_life_arrhenius</code> shall emit a warning when $Ea$ is outside the                                                                                                                                                                       |

|              |                                                    |                                                                                                                                                                                                                                                                                                                                        |
|--------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                                                    | typical range [10, 25] kcal/mol and continue with exit 0.                                                                                                                                                                                                                                                                              |
| UR-SHELF-011 | Arrhenius — temperature direction check            | jrc_shelf_life_arrhenius shall reject inputs where accel_temp ≤ real_temp or an invalid --unit value is supplied, with a non-zero exit.                                                                                                                                                                                                |
| UR-SHELF-012 | Linear — CSV input                                 | jrc_shelf_life_linear shall accept a CSV file with headers 'time' and 'value' containing individual measurements (not aggregated means). At least 3 distinct time points are required.                                                                                                                                                 |
| UR-SHELF-013 | Linear — Brown-Forsythe homogeneity test           | jrc_shelf_life_linear shall perform the Brown-Forsythe test for homogeneity of variance across time points (base R implementation: ANOVA on absolute deviations from group medians). The result is reported; heterogeneity is a warning, not an error.                                                                                 |
| UR-SHELF-014 | Linear — OLS regression and shelf life estimation  | jrc_shelf_life_linear shall fit $\ln(\text{value}) \sim \text{time}$ by OLS and solve for the shelf life as the time at which the lower (or upper, if --direction high) confidence bound of the predicted mean crosses the specification limit.                                                                                        |
| UR-SHELF-015 | Linear — --direction flag                          | jrc_shelf_life_linear shall accept --direction low (default, value must stay above spec) and --direction high (value must stay below spec, growing impurity use case).                                                                                                                                                                 |
| UR-SHELF-016 | Linear — --transform log flag                      | jrc_shelf_life_linear shall accept --transform log, which fits $\ln(\log(\text{value})) \sim \text{time}$ . The confidence interval of the predicted mean is back-transformed via $\exp()$ before comparing to the specification limit. All values must be strictly positive. The model CSV shall record the transform field as 'log'. |
| UR-SHELF-017 | Linear — PNG output                                | jrc_shelf_life_linear shall save a plot PNG to ~/Downloads/ with a timestamp prefix and the suffix _jrc_shelf_life_linear.png.                                                                                                                                                                                                         |
| UR-SHELF-018 | Linear — model CSV output                          | jrc_shelf_life_linear shall save a model coefficient CSV to ~/Downloads/ containing intercept, slope, residual SE, n, t_bar, Sxx, last_time, spec_limit, confidence, direction, transform, and source_file. This file is consumed by jrc_shelf_life_extrapolate.                                                                       |
| UR-SHELF-019 | Linear — reject values below spec at t=0           | jrc_shelf_life_linear shall reject datasets where the observed mean or fit is already below the specification limit at the first time point (direction=low), with a non-zero exit.                                                                                                                                                     |
| UR-SHELF-020 | Poolability — CSV input                            | jrc_shelf_life_poolability shall accept a CSV file with headers 'batch', 'time', and 'value'. At least 2 distinct batches and at least 3 distinct time points per batch are required.                                                                                                                                                  |
| UR-SHELF-021 | Poolability — ICH Q1E two-step ANCOVA              | jrc_shelf_life_poolability shall perform the ICH Q1E two-step ANCOVA at $\alpha=0.25$ : Step 1 tests the batch×time interaction (slope equality); Step 2 tests the batch main effect (intercept equality). Both F-statistics and p-values are reported.                                                                                |
| UR-SHELF-022 | Poolability — pooling decision                     | jrc_shelf_life_poolability shall report one of three decisions: FULL POOL (Step 1 and Step 2 both $p > 0.25$ ), PARTIAL POOL (Step 1 $p > 0.25$ , Step 2 $p \leq 0.25$ ), DO NOT POOL (Step 1 $p \leq 0.25$ ).                                                                                                                         |
| UR-SHELF-023 | Poolability — PNG output                           | jrc_shelf_life_poolability shall save a batch degradation plot PNG to ~/Downloads/ with a timestamp prefix and the suffix _jrc_shelf_life_poolability.png.                                                                                                                                                                             |
| UR-SHELF-024 | Extrapolate — model CSV input and verification     | jrc_shelf_life_extrapolate shall read a model CSV produced by jrc_shelf_life_linear and verify that the 'script' field identifies jrc_shelf_life_linear. Non-existent or wrong-script files shall cause a non-zero exit.                                                                                                               |
| UR-SHELF-025 | Extrapolate — CI projection with transform support | jrc_shelf_life_extrapolate shall compute the fitted value and confidence interval of the mean at the target time using the stored model coefficients. If the model CSV contains transform=log, the CI shall be back-transformed via $\exp()$ before display and comparison to the specification limit.                                 |
| UR-SHELF-026 | Extrapolate — ICH Q1E extrapolation limits         | jrc_shelf_life_extrapolate shall emit a warning when the target time is more than 50% beyond the last observation, and shall reject (non-zero exit) when the target time is more than 100% beyond the last observation.                                                                                                                |

|              |                                |                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UR-SHELF-027 | Extrapolate — input validation | jrc_shelf_life_extrapolate shall reject a negative target_time with a non-zero exit.                                                                                                                                                                                                                                                                                       |
| UR-SHELF-028 | Output — --report DV report    | When invoked with the --report flag, all five Shelf Life scripts shall write a DV report to ~/Downloads/. If jr_pack is installed the report is generated as a .docx file; otherwise the intermediate .html file remains. A JSON sidecar shall contain report_type == "dv" and verdict_pass as a boolean reflecting the overall design verification verdict for the study. |

## 6. Test Environment

| Component        | Specification                                                 |
|------------------|---------------------------------------------------------------|
| Operating System | macOS 15.x (Apple Silicon) or Windows 10/11 (Git Bash)        |
| R Version        | As installed per JR-IQ-001 (R 4.3.x or later)                 |
| Python Version   | As installed per JR-IQ-001 (Python 3.11.x or later)           |
| Test Framework   | pytest 8.x                                                    |
| Test runner      | repos/shelf_life/admin_shelf_life_oq                          |
| Test file        | repos/shelf_life/oq/test_shelf_life_suite.py (72 tests)       |
| Test data        | repos/shelf_life/oq/data/ (13 CSV fixtures)                   |
| OQ venv          | ~/venvs/<PROJECT_ID>_oq/ (shared with other module OQ suites) |
| Output directory | ~/Downloads/ (PNG and model CSV verified by mtime glob)       |
| Entry point      | bin/jrrun (integrity-checked execution)                       |

## 7. Test Cases

All test cases are implemented in repos/shelf\_life/oq/test\_shelf\_life\_suite.py. Test data files reside in repos/shelf\_life/oq/data/. Numerical reference values for correctness tests are computed independently of the R scripts using pure Python arithmetic (math.exp()), OLS in Python, bisection) and NIST t-table values.

### 7.1 jrc\_shelf\_life\_q10

#### TC-SHELF-Q10-001

**Tests:** UR-SHELF-003

**Command:** jrrun jrc\_shelf\_life\_q10.R 2.0 55 25 26

**Rationale:** Valid inputs.

**Pass criterion:** exit 0; 'Acceleration factor', 'Real-time equivalent', 'Sensitivity' in output

#### TC-SHELF-Q10-002

**Tests:** UR-SHELF-003

**Command:** jrrun jrc\_shelf\_life\_q10.R 2.0 55 25 26

**Rationale:** Known values:  $AF = 2.0^{(30/10)} = 8.0$  (exact);  $real\_time = 26 \times 8.0 = 208.0$  (exact). Reference computed by integer arithmetic, independent of R.

**Pass criterion:** exit 0;  $AF = 8.0 \pm 0.001$ ;  $real\_time = 208.0 \pm 0.01$

#### TC-SHELF-Q10-003

**Tests:** UR-SHELF-004

**Command:** jrrun jrc\_shelf\_life\_q10.R 2.0 55 25 26

**Rationale:** Sensitivity table rows for Q10=1.5, Q10=2.0 (stated), Q10=2.5.

**Pass criterion:** exit 0; '1.5', 'stated value', '2.5' in output

#### TC-SHELF-Q10-004

**Tests:** UR-SHELF-005

**Command:** jrrun jrc\_shelf\_life\_q10.R 1.2 55 25 26

**Rationale:** Q10=1.2 (below [1.5, 3.0] typical range).

**Pass criterion:** exit 0; range warning in output

#### TC-SHELF-Q10-005

**Tests:** UR-SHELF-006

**Command:** `jrrun jrc_shelf_life_q10.R 2.0 25 55 26`

**Rationale:**  $\text{accel\_temp}(25) \leq \text{real\_temp}(55)$  — no acceleration.

**Pass criterion:** exit non-zero

#### TC-SHELF-Q10-006

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_q10.R` (no arguments)

**Rationale:** Missing arguments.

**Pass criterion:** exit non-zero or 'Usage' in output

#### TC-SHELF-Q10-007

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_q10.R abc 55 25 26`

**Rationale:** Non-numeric q10.

**Pass criterion:** exit non-zero

#### TC-SHELF-Q10-008

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_q10.R 2.0 55 25 -5`

**Rationale:** Negative accel\_time.

**Pass criterion:** exit non-zero

#### TC-SHELF-Q10-009

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_q10.R --help`

**Rationale:** --help flag.

**Pass criterion:** exit 0; 'Usage' in output

#### TC-SHELF-Q10-010

**Tests:** UR-SHELF-001

**Command:** `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_q10.R 2.0 55 25 26`  
(`RENV_PATHS_ROOT` unset)

**Rationale:** Direct Rscript call bypasses jrrun.

**Pass criterion:** exit non-zero; 'RENV\_PATHS\_ROOT' in output

#### TC-SHELF-Q10-011

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_q10.R 2.0 55 25 26 --report`

**Rationale:** --report flag triggers DV report generation. Accepts .docx (jr\_pack full install) or .html (template-only install).

**Pass criterion:** exit 0; file matching \*\_q10\_dv\_report.docx or \*\_q10\_dv\_report.html found in ~/Downloads/ with recent mtime.

#### TC-SHELF-Q10-012

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_q10.R 2.0 55 25 26 --report`

**Rationale:** JSON sidecar is written alongside the report file.

**Pass criterion:** exit 0; file matching \*\_q10\_dv\_report\_data.json found in ~/Downloads/ with recent mtime.

#### TC-SHELF-Q10-013



**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_q10.R 2.0 55 25 26 --report`

**Rationale:** JSON sidecar must carry `report_type == "dv"` and `verdict_pass == True`. Q10 script always yields `verdict_pass True` (pure calculation, no spec check).

**Pass criterion:** exit 0; JSON found; `report_type == "dv"` and `verdict_pass` is True.

## 7.2 jrc\_shelf\_life\_arrhenius

### TC-SHELF-ARR-001

**Tests:** UR-SHELF-007

**Command:** `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26`

**Rationale:** Valid inputs.

**Pass criterion:** exit 0; 'Acceleration factor', 'Real-time equivalent', 'Sensitivity' in output

### TC-SHELF-ARR-002

**Tests:** UR-SHELF-007

**Command:** `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26`

**Rationale:** Known values:  $AF = \exp(17.0/1.987e-3 \times (1/298.15 - 1/328.15)) = 13.7826$ . Reference computed via `math.exp()` in pure Python, independent of R. Tolerance  $\pm 0.001$ .

**Pass criterion:** exit 0;  $AF = 13.7826 \pm 0.001$ ; `real_time = 358.35 ± 0.05`

### TC-SHELF-ARR-003

**Tests:** UR-SHELF-008

**Command:** `jrrun jrc_shelf_life_arrhenius.R 328.15 298.15 17.0 26 --unit K`

**Rationale:** Kelvin inputs equivalent to 55°C/25°C Celsius run.

**Pass criterion:** exit 0; AF matches Celsius run  $\pm 0.01$

### TC-SHELF-ARR-004

**Tests:** UR-SHELF-009

**Command:** `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26`

**Rationale:** Sensitivity table rows for  $E_a=15.0$ ,  $E_a=17.0$  (stated),  $E_a=19.0$ .

**Pass criterion:** exit 0; '15.0', '17.0', '19.0' in output

### TC-SHELF-ARR-005

**Tests:** UR-SHELF-010

**Command:** `jrrun jrc_shelf_life_arrhenius.R 55 25 30.0 26`

**Rationale:**  $E_a=30.0$  (above [10, 25] typical range).

**Pass criterion:** exit 0; range warning in output

### TC-SHELF-ARR-006

**Tests:** UR-SHELF-011

**Command:** `jrrun jrc_shelf_life_arrhenius.R 25 55 17.0 26`

**Rationale:** `accel_temp (25) ≤ real_temp (55)`.

**Pass criterion:** exit non-zero

### TC-SHELF-ARR-007

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_arrhenius.R` (no arguments)

**Rationale:** Missing arguments.

**Pass criterion:** exit non-zero or 'Usage' in output

### TC-SHELF-ARR-008

**Tests:** UR-SHELF-011

**Command:** `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26 --unit F`

**Rationale:** Invalid --unit value (Fahrenheit not supported).

**Pass criterion:** exit non-zero

#### TC-SHELF-ARR-009

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_arrhenius.R --help`

**Rationale:** --help flag.

**Pass criterion:** exit 0; 'Usage' in output

#### TC-SHELF-ARR-010

**Tests:** UR-SHELF-001

**Command:** `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_arrhenius.R 55 25 17.0 26 (RENV_PATHS_ROOT unset)`

**Rationale:** Direct Rscript call bypasses jrrun.

**Pass criterion:** exit non-zero; 'RENV\_PATHS\_ROOT' in output

#### TC-SHELF-ARR-011

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26 --report`

**Rationale:** --report flag triggers DV report generation. Accepts .docx (jr\_pack full install) or .html (template-only install).

**Pass criterion:** exit 0; file matching \*\_arrhenius\_dv\_report.docx or \*\_arrhenius\_dv\_report.html found in ~/Downloads/ with recent mtime.

#### TC-SHELF-ARR-012

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26 --report`

**Rationale:** JSON sidecar is written alongside the report file.

**Pass criterion:** exit 0; file matching \*\_arrhenius\_dv\_report\_data.json found in ~/Downloads/ with recent mtime.

#### TC-SHELF-ARR-013

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26 --report`

**Rationale:** JSON sidecar must carry report\_type == "dv" and verdict\_pass == True. Arrhenius script always yields verdict\_pass True (pure calculation, no spec check).

**Pass criterion:** exit 0; JSON found; report\_type == "dv" and verdict\_pass is True.

## 7.3 jrc\_shelf\_life\_linear

#### TC-SHELF-LIN-001

**Tests:** UR-SHELF-012, UR-SHELF-013, UR-SHELF-014

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

**Rationale:** Valid 15-row homogeneous dataset (5 time points × 3 units).

**Pass criterion:** exit 0; 'Homogeneity of Variance', 'Regression', 'Shelf Life Estimate' in output

#### TC-SHELF-LIN-002

**Tests:** UR-SHELF-014

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

**Rationale:** Known dataset, spec=80, conf=0.95. Reference shelf life computed by pure-Python OLS + bisection:  $b_0=100.647$ ,  $b_1=-0.807$ ,  $\sigma=0.686$  (hand-verified).  $t\_crit(0.975, df=13)=2.16037$  (NIST t-table). Expected shelf life  $\approx 24.723$ . Tolerance  $\pm 0.05$ .

**Pass criterion:** exit 0; shelf life =  $24.723 \pm 0.05$

#### TC-SHELF-LIN-003

**Tests:** UR-SHELF-013

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

**Rationale:** Homogeneous variance dataset.

**Pass criterion:** exit 0; 'homogeneous' or variance pass in output

#### TC-SHELF-LIN-004

**Tests:** UR-SHELF-013

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_heterogeneous.csv 75.0 0.95`

**Rationale:** Heterogeneous variance dataset.

**Pass criterion:** exit 0; heterogeneous variance warning in output

#### TC-SHELF-LIN-005

**Tests:** UR-SHELF-017

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

**Rationale:** PNG output.

**Pass criterion:** exit 0; \*\_jrc\_shelf\_life\_linear.png found in ~/Downloads/

#### TC-SHELF-LIN-006

**Tests:** UR-SHELF-018

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

**Rationale:** Model CSV output.

**Pass criterion:** exit 0; \*\_jrc\_shelf\_life\_linear\_model.csv found in ~/Downloads/

#### TC-SHELF-LIN-007

**Tests:** UR-SHELF-015

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_direction_high.csv 0.20 0.95 --direction high`

**Rationale:** --direction high (growing impurity data).

**Pass criterion:** exit 0; 'upper' or 'high' in output

#### TC-SHELF-LIN-008

**Tests:** UR-SHELF-019

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_below_spec.csv 80.0 0.95`

**Rationale:** Values already below spec at t=0 (direction=low).

**Pass criterion:** exit non-zero

#### TC-SHELF-LIN-009

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_linear.R (no arguments)`

**Rationale:** Missing arguments.

**Pass criterion:** exit non-zero or 'Usage' in output

#### TC-SHELF-LIN-010

**Tests:** UR-SHELF-012

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_missing_col.csv 80.0 0.95`

**Rationale:** CSV missing 'value' column.

**Pass criterion:** exit non-zero; 'value' in error output

#### TC-SHELF-LIN-011

**Tests:** UR-SHELF-012

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_few_timepoints.csv 90.0 0.95`

**Rationale:** Only 2 distinct time points (< 3 required).

**Pass criterion:** exit non-zero

#### TC-SHELF-LIN-012

**Tests:** UR-SHELF-001

**Command:** `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 (RENV_PATHS_ROOT unset)`

**Rationale:** Direct Rscript call bypasses jrrun.

**Pass criterion:** exit non-zero; 'RENV\_PATHS\_ROOT' in output

#### TC-SHELF-LIN-013

**Tests:** UR-SHELF-016

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --transform log`

**Rationale:** --transform log flag; log transform noted in output.

**Pass criterion:** exit 0; 'log' in output; 'Shelf Life Estimate' in output

#### TC-SHELF-LIN-014

**Tests:** UR-SHELF-016

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --transform log`

**Rationale:** Known log-linear shelf life: OLS on  $\log(\text{value}) \sim \text{time}$  computed in pure Python. CI back-transformed via `exp()`. Reference shelf life computed via bisection. Tolerance  $\pm 0.05$ .

**Pass criterion:** exit 0; shelf life = `_SHELF_LIFE_LOG_EXPECTED`  $\pm 0.05$

#### TC-SHELF-LIN-015

**Tests:** UR-SHELF-016

**Command:** `jrrun jrc_shelf_life_linear.R <data with value=0> 70.0 0.95 --transform log`

**Rationale:** Data contains value  $\leq 0$  with --transform log.

**Pass criterion:** exit non-zero; 'positive' or ' $\leq 0$ ' in error output

#### TC-SHELF-LIN-016

**Tests:** UR-SHELF-016, UR-SHELF-018

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --transform log`

**Rationale:** Model CSV must record transform field.

**Pass criterion:** exit 0; model CSV contains 'transform' = 'log'

#### TC-SHELF-LIN-017

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --report`

**Rationale:** --report flag triggers DV report generation. Accepts .docx (jr\_pack full install) or .html (template-only install).

**Pass criterion:** exit 0; file matching `*_shelf_life_linear_dv_report.docx` or `*_shelf_life_linear_dv_report.html` found in `~/Downloads/` with recent mtime.

#### TC-SHELF-LIN-018

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --report`

**Rationale:** JSON sidecar is written alongside the report file.

**Pass criterion:** exit 0; file matching \*\_shelf\_life\_linear\_dv\_report\_data.json found in ~/Downloads/ with recent mtime.

#### TC-SHELF-LIN-019

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --report`

**Rationale:** JSON sidecar must carry report\_type == "dv" and verdict\_pass == True for passing data. TC-SHELF-LIN-002 confirms shelf life > 0 for homogeneous.csv, spec=80.

**Pass criterion:** exit 0; JSON found; report\_type == "dv" and verdict\_pass is True.

## 7.4 jrc\_shelf\_life\_poolability

#### TC-SHELF-POOL-001

**Tests:** UR-SHELF-020, UR-SHELF-021, UR-SHELF-022

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv`

**Rationale:** Valid 3-batch dataset.

**Pass criterion:** exit 0; 'Decision', 'POOL', 'Step 1', 'Step 2' in output

#### TC-SHELF-POOL-002

**Tests:** UR-SHELF-022

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv`

**Rationale:** Dataset with equal slopes and intercepts. Both ANCOVA steps  $p > 0.25$ .

**Pass criterion:** exit 0; 'FULL POOL' in output; Step 1  $p > 0.25$ ; Step 2  $p > 0.25$

#### TC-SHELF-POOL-003

**Tests:** UR-SHELF-022

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_no_pool.csv`

**Rationale:** Dataset with clearly different slopes (A=-0.4, B=-0.8, C=-1.2 per unit time). Step 1 interaction must be highly significant ( $p < 0.05$ ).

**Pass criterion:** exit 0; 'DO NOT POOL' in output; Step 1  $p < 0.05$

#### TC-SHELF-POOL-004

**Tests:** UR-SHELF-022

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_partial.csv`

**Rationale:** Dataset with equal slopes and different intercepts. Step 1  $p > 0.25$ ; Step 2  $p < 0.01$ .

**Pass criterion:** exit 0; 'PARTIAL POOL' in output; Step 1  $p > 0.25$ ; Step 2  $p < 0.01$

#### TC-SHELF-POOL-005

**Tests:** UR-SHELF-023

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv`

**Rationale:** PNG output.

**Pass criterion:** exit 0; \*\_jrc\_shelf\_life\_poolability.png found in ~/Downloads/

#### TC-SHELF-POOL-006

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_poolability.R` (no arguments)

**Rationale:** Missing arguments.

**Pass criterion:** exit non-zero or 'Usage' in output

#### TC-SHELF-POOL-007

**Tests:** UR-SHELF-020

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_missing_col.csv`

**Rationale:** CSV missing 'batch' column.

**Pass criterion:** exit non-zero; 'batch' in error output

#### TC-SHELF-POOL-008

**Tests:** UR-SHELF-020

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_one_batch.csv`

**Rationale:** Only 1 batch (< 2 required).

**Pass criterion:** exit non-zero

#### TC-SHELF-POOL-009

**Tests:** UR-SHELF-020

**Command:** `jrrun jrc_shelf_life_poolability.R <2-timepoint CSV>`

**Rationale:** Fewer than 3 distinct time points per batch.

**Pass criterion:** exit non-zero

#### TC-SHELF-POOL-010

**Tests:** UR-SHELF-001

**Command:** `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv (RENV_PATHS_ROOT unset)`

**Rationale:** Direct Rscript call bypasses jrrun.

**Pass criterion:** exit non-zero; 'RENV\_PATHS\_ROOT' in output

#### TC-SHELF-POOL-011

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv --report`

**Rationale:** --report flag triggers DV report generation. Accepts .docx (jr\_pack full install) or .html (template-only install).

**Pass criterion:** exit 0; file matching \*\_poolability\_dv\_report.docx or \*\_poolability\_dv\_report.html found in ~/Downloads/ with recent mtime.

#### TC-SHELF-POOL-012

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv --report`

**Rationale:** JSON sidecar is written alongside the report file.

**Pass criterion:** exit 0; file matching \*\_poolability\_dv\_report\_data.json found in ~/Downloads/ with recent mtime.

#### TC-SHELF-POOL-013

**Tests:** UR-SHELF-028

**Command:** `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv --report`

**Rationale:** JSON sidecar must carry report\_type == "dv" and verdict\_pass == True for poolable data. TC-SHELF-POOL-002 confirms FULL POOL for pool\_poolable.csv.

**Pass criterion:** exit 0; JSON found; report\_type == "dv" and verdict\_pass is True.

## 7.5 jrc\_shelf\_life\_extrapolate

#### TC-SHELF-EXT-001

**Tests:** UR-SHELF-024, UR-SHELF-025

**Command:** `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 20`

**Rationale:** Valid model CSV and target within observed range.

**Pass criterion:** exit 0; 'Projection', 'Fitted value' in output

#### TC-SHELF-EXT-002

**Tests:** UR-SHELF-025

**Command:** `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 20`

**Rationale:** Known numerical check (target=20):  $\text{fit} = 100.647 - 0.807 \times 20 = 84.502$  ( $\pm 0.01$ ). CI lower = 83.976 ( $\pm 0.05$ ) from closed-form formula with  $t_{\text{crit}}(0.975, \text{df}=13)=2.16037$  (NIST t-table). References computed independently of R.

**Pass criterion:** exit 0; fitted value =  $84.502 \pm 0.01$ ; lower 95% CI bound =  $83.976 \pm 0.05$

### TC-SHELF-EXT-003

**Tests:** UR-SHELF-025

**Command:** `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 20`

**Rationale:** Lower CI > spec=80 at target=20.

**Pass criterion:** exit 0; 'supported' or PASS message in output

### TC-SHELF-EXT-004

**Tests:** UR-SHELF-025

**Command:** `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 30`

**Rationale:** Lower CI < spec=80 at target=30 (beyond shelf life).

**Pass criterion:** exit 0; 'NOT supported' or 'crossed' in output

### TC-SHELF-EXT-005

**Tests:** UR-SHELF-026

**Command:** `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 37`

**Rationale:** Target=37 is 54% beyond last\_time=24 (> 50% warning threshold).

**Pass criterion:** exit 0; extrapolation warning ('beyond', 'wide', or '50') in output

### TC-SHELF-EXT-006

**Tests:** UR-SHELF-026

**Command:** `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 50`

**Rationale:** Target=50 is 108% beyond last\_time=24 (> 100% hard stop).

**Pass criterion:** exit non-zero; '100', 'rejected', or 'speculative' in output

### TC-SHELF-EXT-007

**Tests:** UR-SHELF-002

**Command:** `jrrun jrc_shelf_life_extrapolate.R` (no arguments)

**Rationale:** Missing arguments.

**Pass criterion:** exit non-zero or 'Usage' in output

### TC-SHELF-EXT-008

**Tests:** UR-SHELF-024

**Command:** `jrrun jrc_shelf_life_extrapolate.R /tmp/no_such_model.csv 20`

**Rationale:** Non-existent model file.

**Pass criterion:** exit non-zero; 'not found' in output

### TC-SHELF-EXT-009

**Tests:** UR-SHELF-024

**Command:** `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_wrongscript.csv 20`

**Rationale:** Model file from wrong script (script field does not contain jrc\_shelf\_life\_linear).

**Pass criterion:** exit non-zero; 'script' in error output

### TC-SHELF-EXT-010

**Tests:** UR-SHELF-027

**Command:** `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv -5`

**Rationale:** Negative target\_time.

**Pass criterion:** exit non-zero

### TC-SHELF-EXT-011

**Tests:** UR-SHELF-001

**Command:** Rscript --vanilla repos/shelf\_life/R/jrc\_shelf\_life\_extrapolate.R shelf\_life\_extrapolate\_model.csv 20 (RENV\_PATHS\_ROOT unset)

**Rationale:** Direct Rscript call bypasses jrrun.

**Pass criterion:** exit non-zero; 'RENV\_PATHS\_ROOT' in output

### TC-SHELF-EXT-012

**Tests:** UR-SHELF-028

**Command:** jrrun jrc\_shelf\_life\_extrapolate.R shelf\_life\_extrapolate\_model.csv 20 -  
-report

**Rationale:** --report flag triggers DV report generation. Accepts .docx (jr\_pack full install) or .html (template-only install).

**Pass criterion:** exit 0; file matching \*\_extrapolate\_dv\_report.docx or \*\_extrapolate\_dv\_report.html found in ~/Downloads/ with recent mtime.

### TC-SHELF-EXT-013

**Tests:** UR-SHELF-028

**Command:** jrrun jrc\_shelf\_life\_extrapolate.R shelf\_life\_extrapolate\_model.csv 20 -  
-report

**Rationale:** JSON sidecar is written alongside the report file.

**Pass criterion:** exit 0; file matching \*\_extrapolate\_dv\_report\_data.json found in ~/Downloads/ with recent mtime.

### TC-SHELF-EXT-014

**Tests:** UR-SHELF-028

**Command:** jrrun jrc\_shelf\_life\_extrapolate.R shelf\_life\_extrapolate\_model.csv 20 -  
-report

**Rationale:** JSON sidecar must carry report\_type == "dv" and verdict\_pass == True. TC-SHELF-EXT-003 confirms CI > spec=80 at target=20.

**Pass criterion:** exit 0; JSON found; report\_type == "dv" and verdict\_pass is True.

## 8. Requirements Traceability Matrix

Each user requirement is covered by at least one test case.

| UR ID        | Title                         | Test Cases                                                                                                             |
|--------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------|
| UR-SHELF-001 | Bypass protection             | TC-SHELF-Q10-010, TC-SHELF-ARR-010, TC-SHELF-LIN-012, TC-SHELF-POOL-010, TC-SHELF-EXT-011                              |
| UR-SHELF-002 | Non-zero exit on error        | TC-SHELF-Q10-005..009, TC-SHELF-ARR-006..009, TC-SHELF-LIN-008..011, TC-SHELF-POOL-006..009, TC-SHELF-EXT-007, EXT-010 |
| UR-SHELF-003 | Q10 — AF computation          | TC-SHELF-Q10-001, TC-SHELF-Q10-002                                                                                     |
| UR-SHELF-004 | Q10 — sensitivity table       | TC-SHELF-Q10-003                                                                                                       |
| UR-SHELF-005 | Q10 — range warning           | TC-SHELF-Q10-004                                                                                                       |
| UR-SHELF-006 | Q10 — temp direction check    | TC-SHELF-Q10-005                                                                                                       |
| UR-SHELF-007 | Arrhenius — AF computation    | TC-SHELF-ARR-001, TC-SHELF-ARR-002                                                                                     |
| UR-SHELF-008 | Arrhenius — unit flag         | TC-SHELF-ARR-003                                                                                                       |
| UR-SHELF-009 | Arrhenius — sensitivity table | TC-SHELF-ARR-004                                                                                                       |
| UR-SHELF-010 | Arrhenius — Ea range warning  | TC-SHELF-ARR-005                                                                                                       |
| UR-SHELF-011 | Arrhenius — temp/unit check   | TC-SHELF-ARR-006, TC-SHELF-ARR-008                                                                                     |
| UR-SHELF-012 | Linear — CSV input            | TC-SHELF-LIN-001, TC-SHELF-LIN-010, TC-SHELF-LIN-011                                                                   |
| UR-SHELF-013 | Linear — Brown-Forsythe test  | TC-SHELF-LIN-003, TC-SHELF-LIN-004                                                                                     |
| UR-SHELF-014 | Linear — OLS + shelf life     | TC-SHELF-LIN-001, TC-SHELF-LIN-002                                                                                     |
| UR-SHELF-015 | Linear — --direction flag     | TC-SHELF-LIN-007                                                                                                       |
| UR-SHELF-016 | Linear — --transform log flag | TC-SHELF-LIN-013, TC-SHELF-LIN-014, TC-SHELF-LIN-015, TC-                                                              |



|              |                                   |                                                                                                                    |
|--------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|
|              |                                   | SHELF-LIN-016                                                                                                      |
| UR-SHELF-017 | Linear — PNG output               | TC-SHELF-LIN-005                                                                                                   |
| UR-SHELF-018 | Linear — model CSV output         | TC-SHELF-LIN-006, TC-SHELF-LIN-016                                                                                 |
| UR-SHELF-019 | Linear — reject below spec at t=0 | TC-SHELF-LIN-008                                                                                                   |
| UR-SHELF-020 | Poolability — CSV input           | TC-SHELF-POOL-001, TC-SHELF-POOL-007, TC-SHELF-POOL-008, TC-SHELF-POOL-009                                         |
| UR-SHELF-021 | Poolability — ANCOVA              | TC-SHELF-POOL-001, TC-SHELF-POOL-002, TC-SHELF-POOL-003, TC-SHELF-POOL-004                                         |
| UR-SHELF-022 | Poolability — pooling decision    | TC-SHELF-POOL-002, TC-SHELF-POOL-003, TC-SHELF-POOL-004                                                            |
| UR-SHELF-023 | Poolability — PNG output          | TC-SHELF-POOL-005                                                                                                  |
| UR-SHELF-024 | Extrapolate — model CSV + verify  | TC-SHELF-EXT-001, TC-SHELF-EXT-008, TC-SHELF-EXT-009                                                               |
| UR-SHELF-025 | Extrapolate — CI projection       | TC-SHELF-EXT-001, TC-SHELF-EXT-002, TC-SHELF-EXT-003, TC-SHELF-EXT-004                                             |
| UR-SHELF-026 | Extrapolate — ICH Q1E limits      | TC-SHELF-EXT-005, TC-SHELF-EXT-006                                                                                 |
| UR-SHELF-027 | Extrapolate — negative time check | TC-SHELF-EXT-010                                                                                                   |
| UR-SHELF-028 | All scripts — --report DV report  | TC-SHELF-Q10-011..013, TC-SHELF-ARR-011..013, TC-SHELF-LIN-017..019, TC-SHELF-POOL-011..013, TC-SHELF-EXT-012..014 |

## 9. Acceptance Criteria

The Shelf Life & Degradation Analysis module OQ is considered PASS when all of the following conditions are satisfied:

- All 72 automated test cases execute with the result PASSED.
- Zero test cases result in FAILED or ERROR.
- The OQ evidence file is generated and preserved at  
~/.jrscript/<PROJECT\_ID>/validation/shelf\_life\_oq\_execution\_<timestamp>.txt.
- All test cases are executed using the validated jrrun entry-point (bypass protection verified for all 5 scripts).
- No open deviations remain unresolved at the time of approval.

Any test failure shall be investigated, documented as a deviation, resolved, and the affected test(s) re-executed. The OQ may not be closed until all deviations are resolved.